STAT 387 **HOMEWORK # 1** Winter 2026 INTRODUCTION TO STATISTICAL LEARNING

DUE DATE: **January 14, 11:59 pm**

Instructions:

- *Please show all the work with proper mathematical/statistical notation and logical flow to receive full points.*
- *You are allowed to use **R** or a Scientific Calculator where required.*
- *Submit only **one*** **PDF** *on* CANVAS.

All questions here are taken from the Exercises pages in the book for Chapter 2 (**pages 52 – 57**): Introduction to Statistical Learning with Applications in R

Conceptual

1. 2 (a, b and c)

a) This is a regression problem because the response variable, CEO salary, is quantitative. We are primarily interested in inference, since the goal is to understand which factors influence CEO salary rather than to make predictions for new firms.
The sample size is n = 500, corresponding to the 500 firms. The number of predictors is p = 3: profit, number of employees, and industry.
b) This is a classification problem because the response variable indicates success or failure, which is categorical. We are most interested in prediction, since the

objective is to predict whether a newly launched product will be successful.
The sample size is n = 20, representing the 20 previously launched products. The number of predictors is p = 14, consisting of price, marketing budget, competition price, and ten additional variables.

c) This is a regression problem because the response variable, the percentage change in the US dollar, is continuous. We are primarily interested in prediction, as the goal is to predict changes in the dollar based on movements in global stock markets.
The sample size is n = 52, corresponding to the 52 weeks in 2012. The number of predictors is p = 3: the percentage change in the US, British, and German markets.

2. 4 (a and b)

a)1) Determining whether a specific credit card transaction is fraudulent. The response variable is binary: fraudulent vs. legitimate. Predictors could include transaction amount, time of day, merchant type, and location. This is a prediction problem because the goal is to flag and prevent fraudulent charges in real-time with high accuracy, rather than primarily understanding the underlying relationship between the predictors and fraud.
a)2) Medical Diagnosis is a classic classification application. The response variable is categorical, such as positive vs. negative (or malignant vs. benign). Predictors include cell size, shape, uniformity, patient age, and genetic markers. This is primarily a prediction model because the immediate goal is to accurately classify the patient's condition to determine the necessary treatment, prioritizing diagnostic accuracy over interpretability.
a)3) Determining student success to identify risk factors. The response variable is categorical: pass vs. fail. Predictors include study time, GPA, class size, and teacher tenure. This would be an inference model because the goal is to understand the relationship between the predictors and the outcome—specifically, to determine which

factors (like class size or study time) have the most significant impact on passing rates to improve curriculum or identify root causes.

b)1) Using regression to predict something like coffee quality could be beneficial. If we had a response variable for quality that was on a scale of 1-10, and predictors such as country of origin, pH levels, moisture content, processing methods, bean color and any other industry relevant predictors that we can collect. The goal is inference because we want to quantify the relationship between predictors like soil pH and the quality score to understand how to improve the farming process.
b)2) Air quality index would be a very beneficial regression application, we could use Air Quality Index as a response variable, and use predictors such as traffic, wind speed, humidity, and the proximity to industry. This would be a prediction application to provide daily air quality forecasts to warn citizens of health risks that are more present at certain times of the day or season.
b)3) Real Estate Pricing. The response variable is House Price. Predictors include zip code, square footage, and number of bedrooms. This is a prediction application. The goal is to input a specific home's features to accurately predict its selling price, helping a seller set the right listing price or a buyer know if they are overpaying.

Applied

3. 9 (a, b, c, d, e, f)

a)Quantitative: mpg, displacement, horsepower, weight, acceleration
Qualitative: Origin, Name, cylinders, year

b)Range of Predictors
MPG Range: (9.0, 46.6)
Displacement Range: (68, 455)
Horsepower Range: (46, 230)
Weight Range: (1613, 5140)
Acceleration Range: (8.0, 24.8)
Year Range: (70, 82)

c) Mean and Standard deviation of predictors
MPG Mean:23.44592
MPG Standard Deviation: 7.805007

Displacement Mean: 194.412
Displacement Standard Deviation: 104.644

Horsepower Mean: 104.4694
Horsepower Standard Deviation: 38.49116

Weight Mean: 2977.584
Weight Standard Deviation: 849.4026

Acceleration Mean: 15.54133
Acceleration Standard Deviation: 2.758864

d)Mean, standard deviation and range for subset of data

MPG Range:(11.0-46.6)
MPG Mean:24.40443
MPG Standard Deviation: 7.867283

Displacement Range: (68, 455)
Displacement Mean: 187.2405
Displacement Standard Deviation: 99.67837

Horsepower Range: (46, 230)
Horsepower Mean: 100.7215
Horsepower Standard Deviation: 35.70885

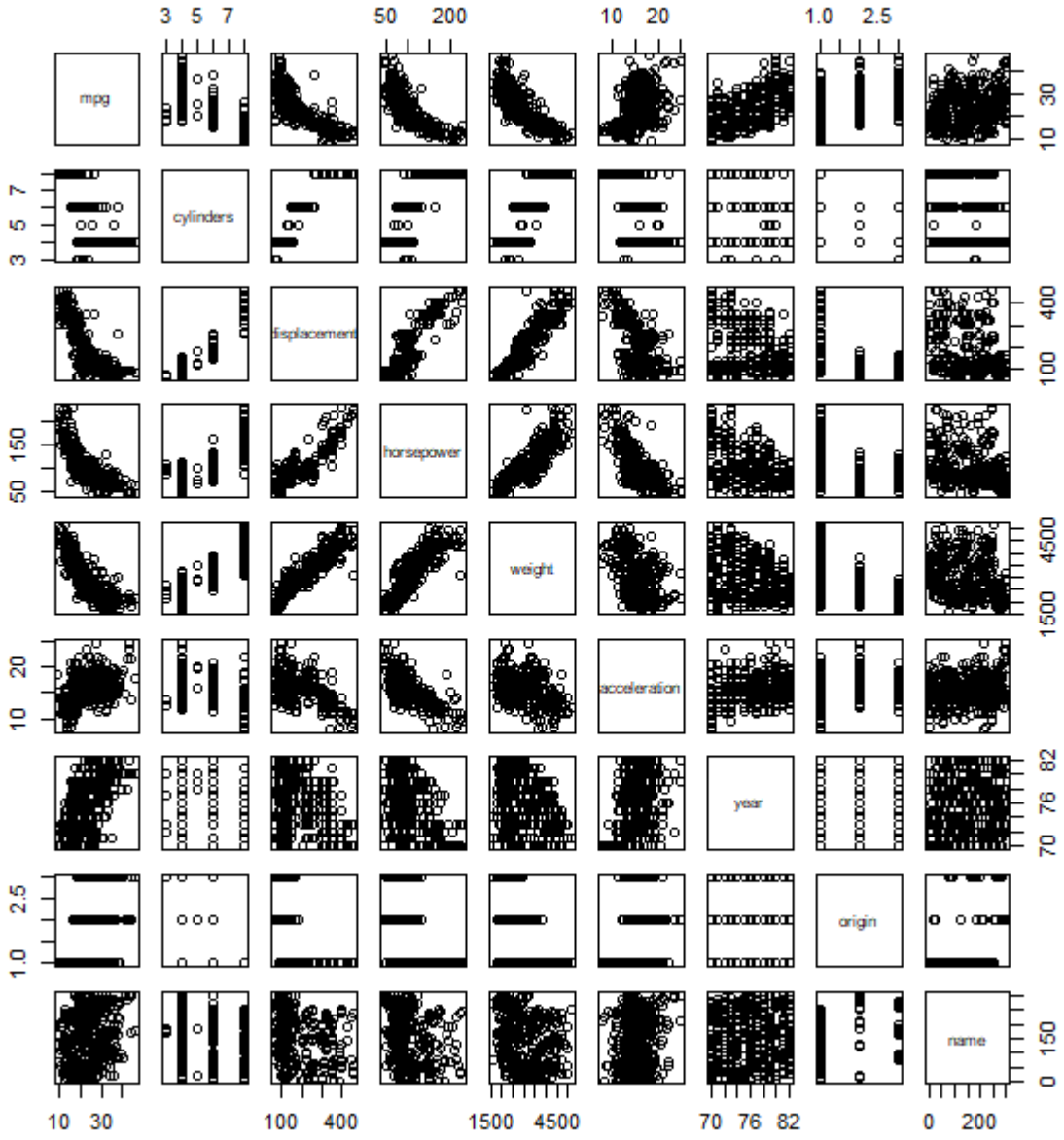
Weight Range: (1649, 4997)
Weight Mean: 2935.972
Weight Standard Deviation: 811.3002

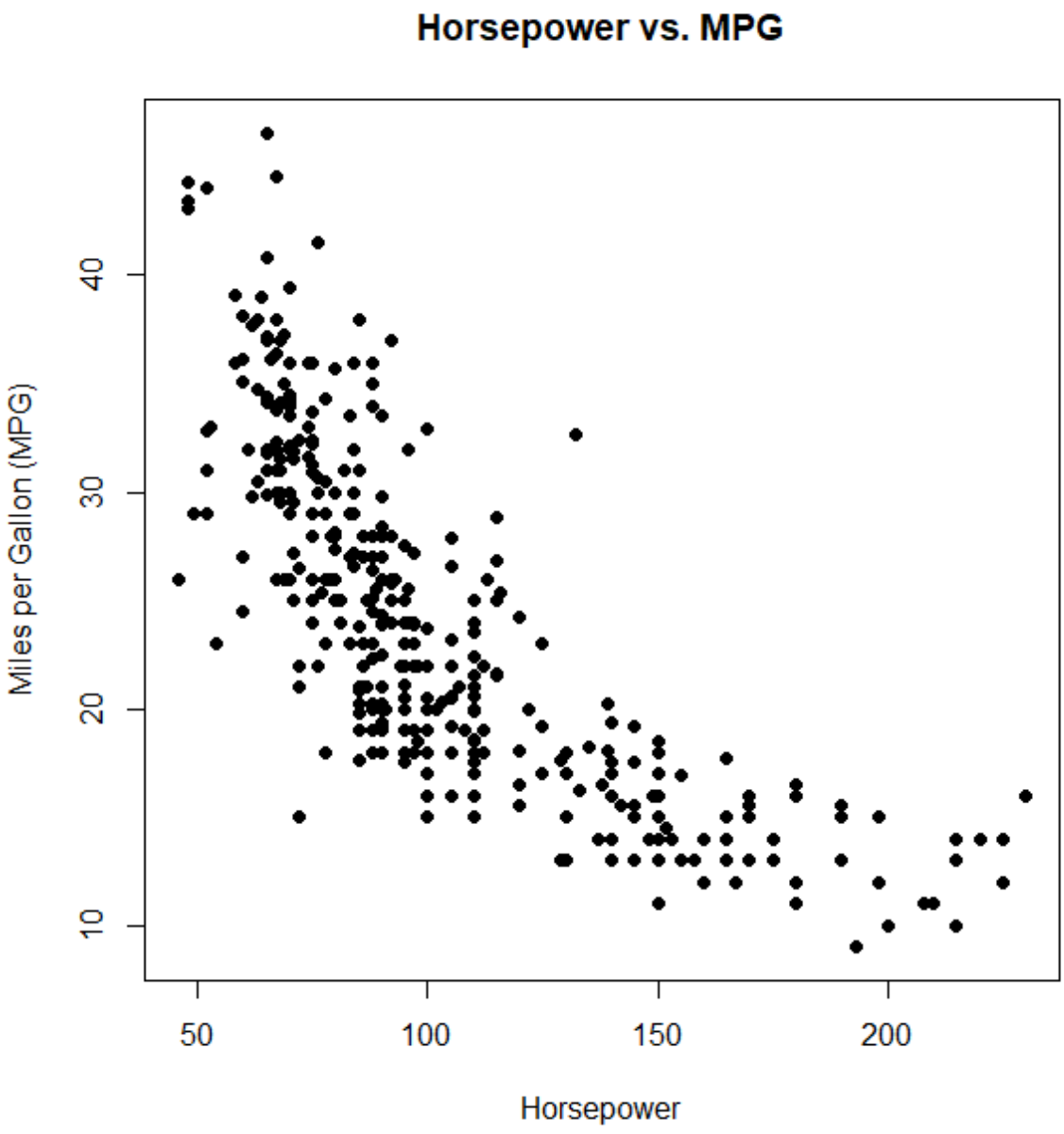
Acceleration Range: (8.5, 24.8)
Acceleration Mean: 15.7269
Acceleration Standard Deviation: 2.693721

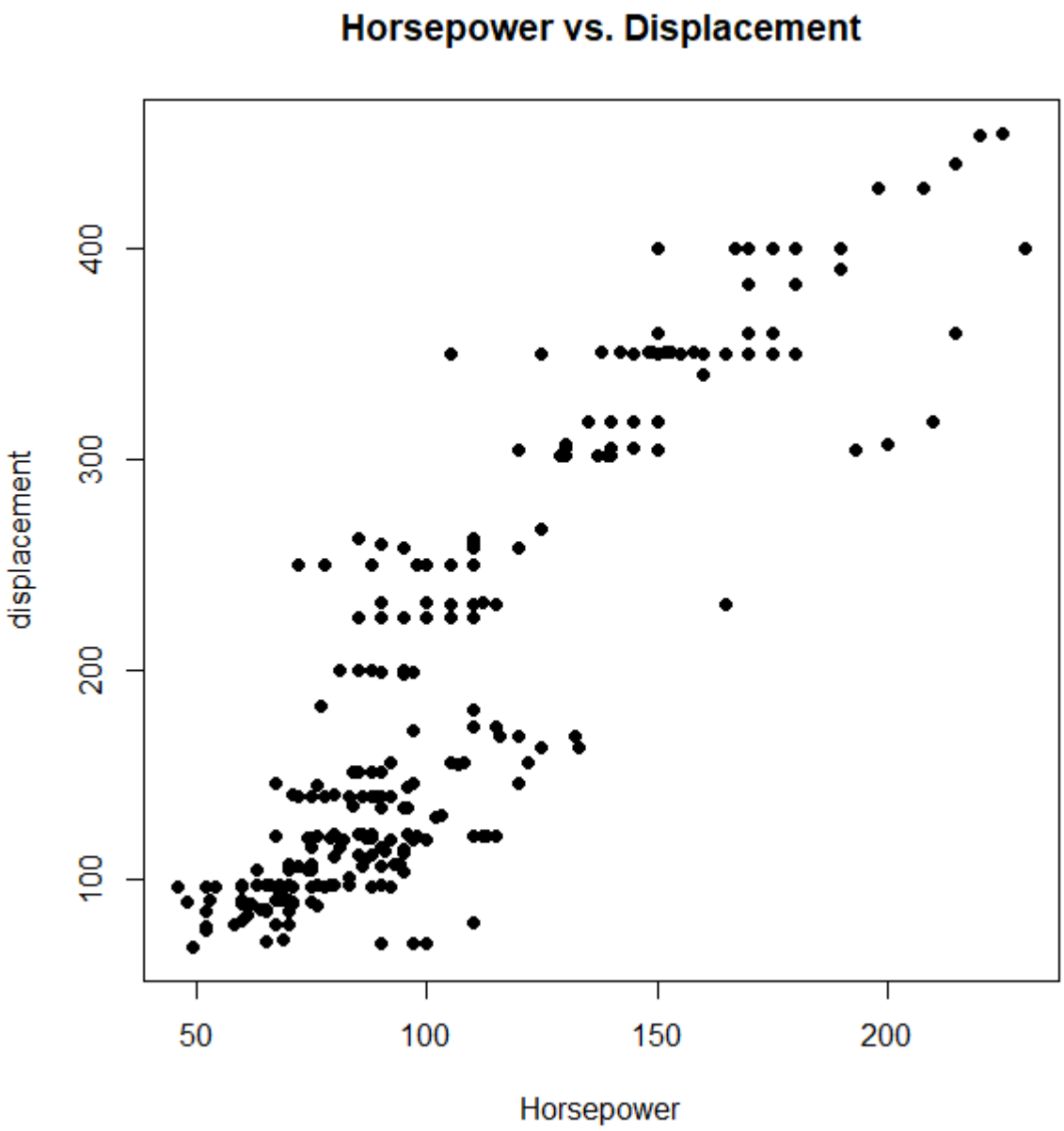
e) Plots are below comments.

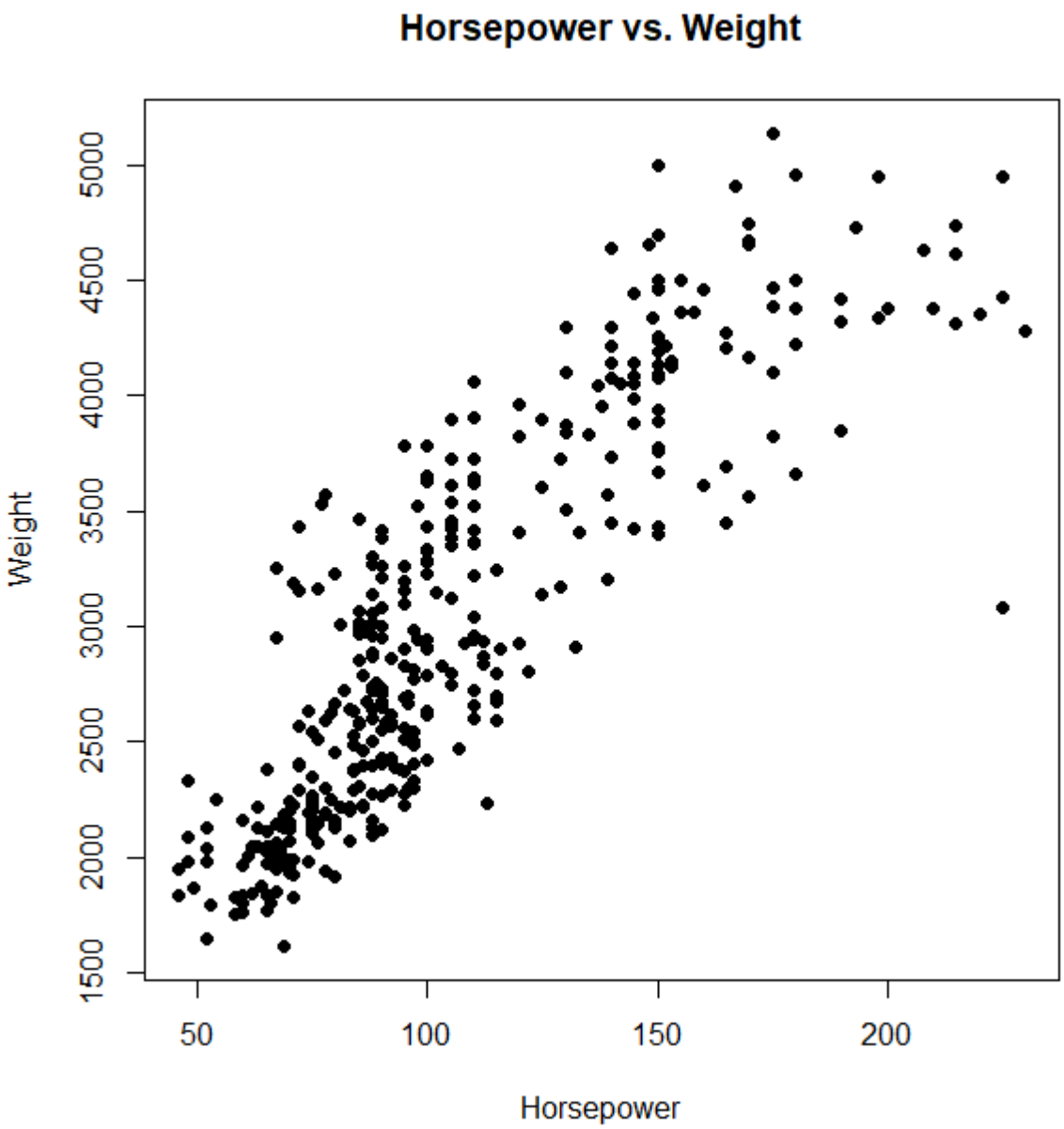
For this part, I was curious how horsepower and different variables interacted. In most of the data, as horsepower goes up, so does weight and displacement, but acceleration time falls and MPG falls in relation to these other predictors increasing. This makes some sense practically, as a large engine is needed to get more horsepower, but that also means more weight is required to make use

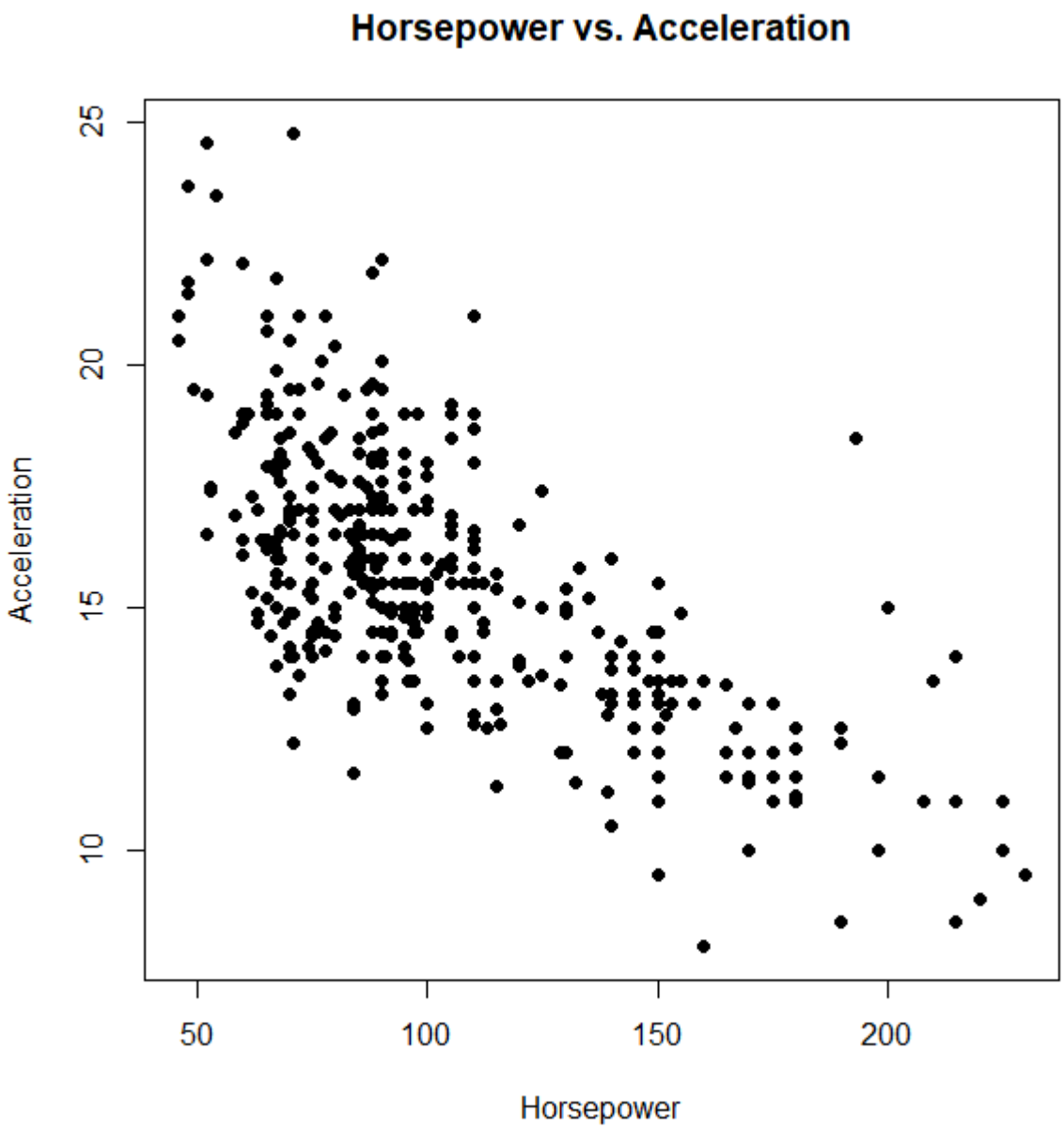
of the bigger engine. I would be interested in investigating this a little further to see if a vehicle that is lighter with a smaller engine would have similar relative horsepower and performance to a larger engine on a heavier vehicle.



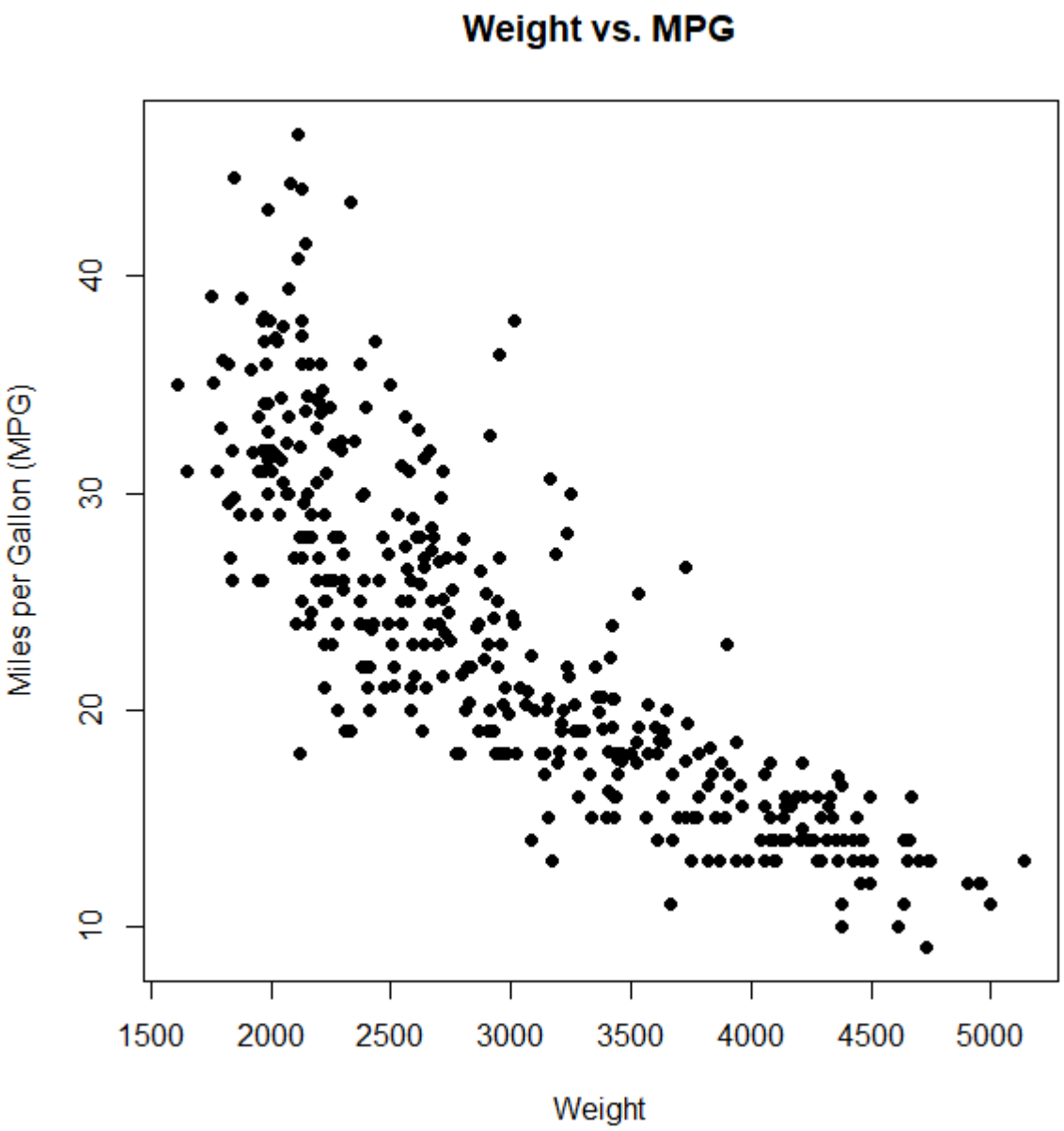

Horsepower vs. MPG
Miles per Gallon (MPG)
10
20
30
40
50
100
150
200
Horsepower

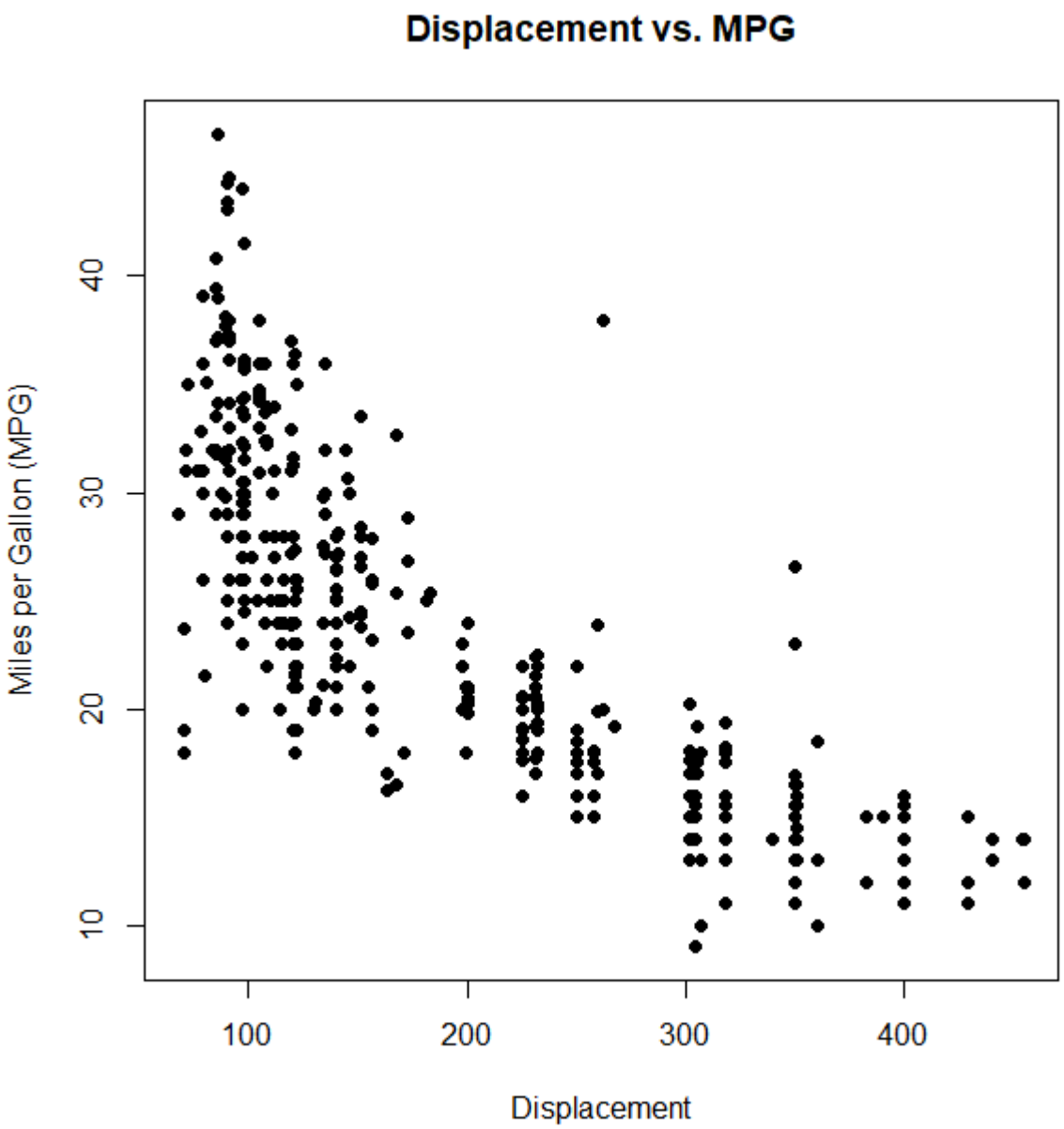

Horsepower vs. Displacement
displacement
100
200
300
400
50
100
150
200
Horsepower

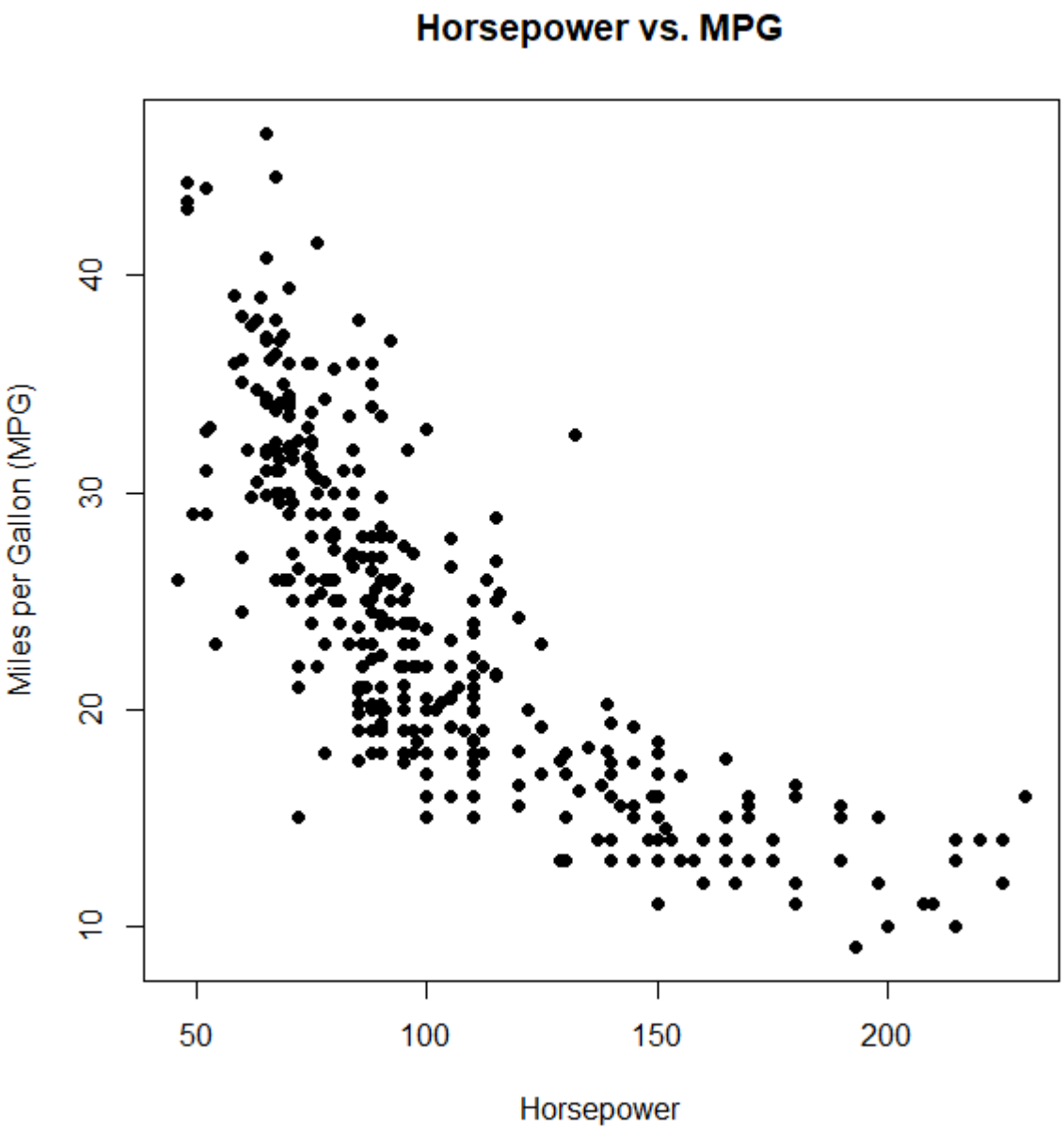

Horsepower vs. Weight
Weight
1500
2000
2500
3000
3500
4000
4500
5000
50
100
150
200
Horsepower

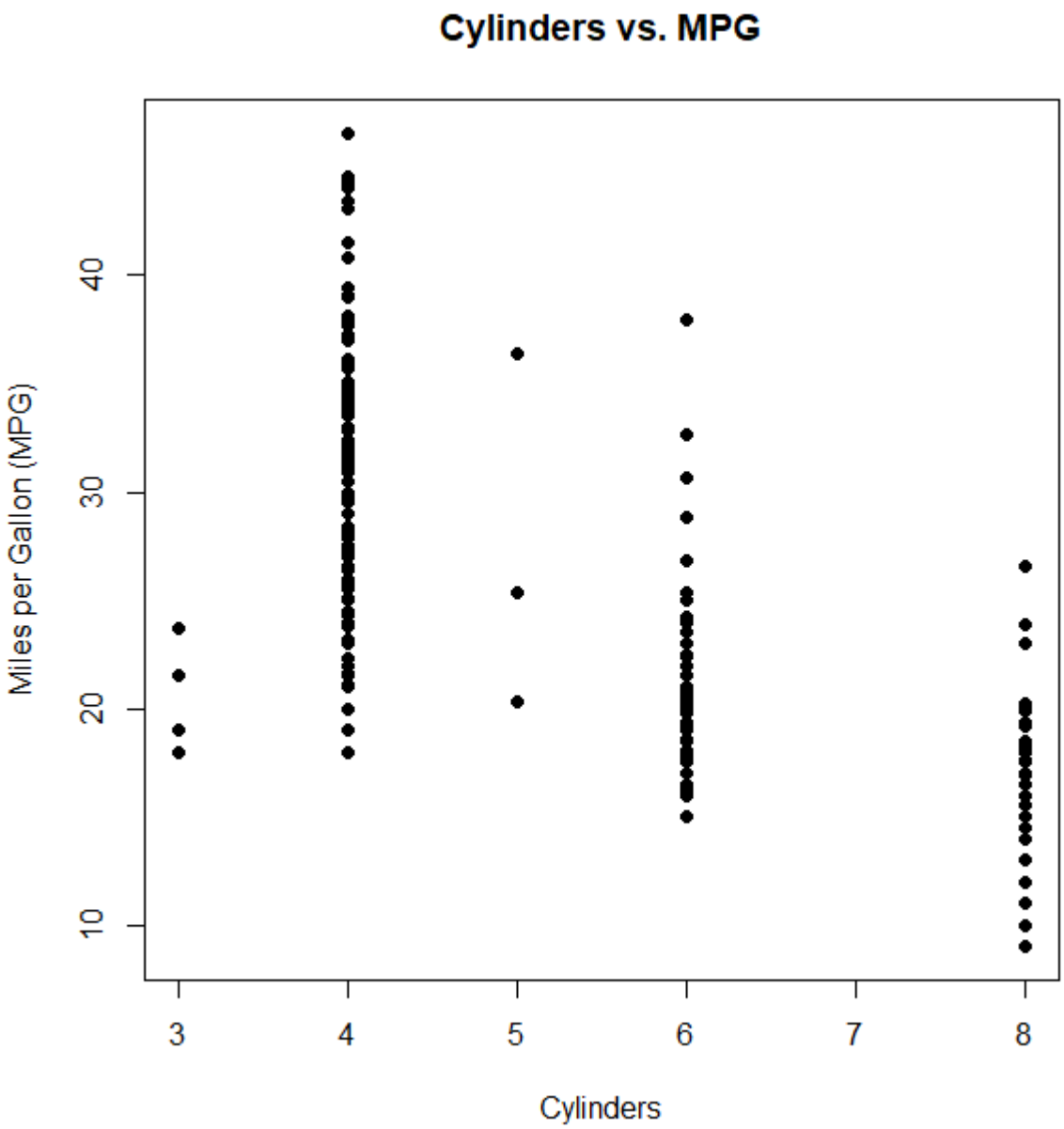

Horsepower vs. Acceleration
Acceleration
25
20
15
10
50
100
150
200
Horsepower

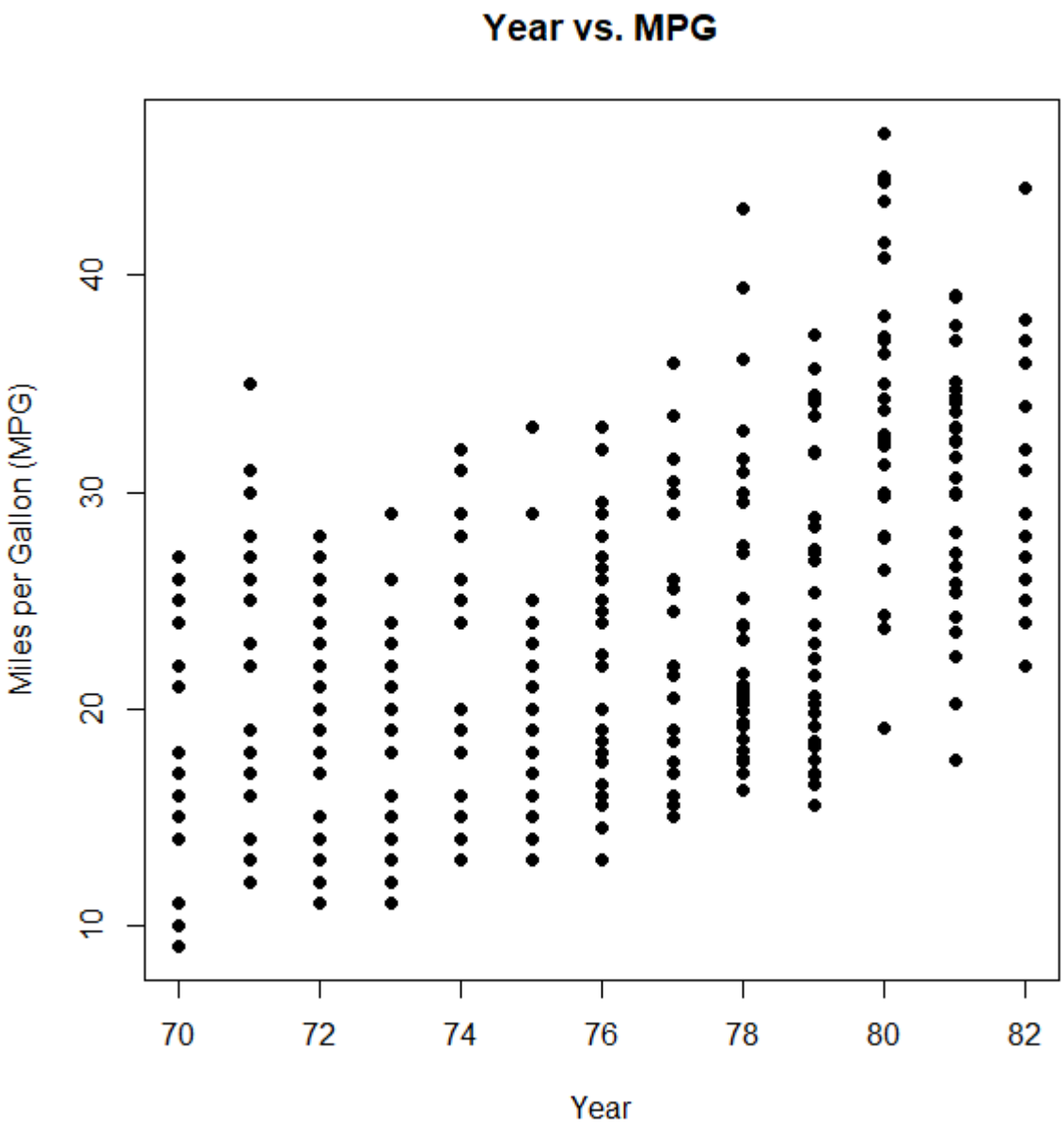
f)I would say that these plots suggest a strong, structured relationship between MPG and weight, displacement, horsepower, year, and cylinders. All five of these predictors show a decent spread across the plots that suggests a linear relationship is present. Investigating these variables makes sense, since a heavier vehicle requires more fuel, and an engine with more horsepower requires more fuel to provide that power. If the engine has more cylinders, that is usually for performance, so fuel consumption will be higher; similarly, a larger displacement means a larger amount of fuel is consumed by the engine. Year is also a major factor, with changes to technology and fuel efficiency increasing the average MPG year after year.


Weight vs. MPG
Miles per Gallon (MPG)
10
20
30
40
1500
2000
2500
3000
3500
4000
4500
5000
Weight


Displacement vs. MPG
Miles per Gallon (MPG)
10
20
30
40
100
200
300
400
Displacement


Horsepower vs. MPG
Miles per Gallon (MPG)
10
20
30
40
50
100
150
200
Horsepower


Cylinders vs. MPG
Miles per Gallon (MPG)
10
20
30
40
3
4
5
6
7
8
Cylinders


Year vs. MPG
Miles per Gallon (MPG)
10
20
30
40
70
72
74
76
78
80
82
Year